

17-Python Assignment Q&A Session

Introduction

- Transcript of a programming class discussion led by instructor Abhishek, including student questions and detailed explanations regarding Python assignments.

Assignment Q&A

- Addressed questions across various assignments: integers, floats, complex numbers, Boolean logic, strings, and loops.
- Clarified usage of bitwise operators in Boolean assignments (e.g., result of left-shifting and right-shifting values).
- Discussed proper string slicing practices, especially avoiding 0 or -1 for certain tasks and implications of slicing index choices.
- Clarified expectation for using loops vs. built-in methods for tasks like finding character counts or indexes; some questions required loops even if string methods seemed available.
- Provided examples on how to find all occurrences of a character/substring using loops, including edge cases and approaches using `index` and `find` methods, with suggestions for try-except and conditional checks to handle errors or terminate loops.

Python Concepts and Methods

- Demonstrated string manipulation: slicing, counting, finding, and replacing elements.
- Provided code walkthroughs for:

- Right/left justification using string methods (`ljust`, `rjust`) and manual implementation with loops.
- Splitting strings into words manually using loops (not using built-in `split` method where not allowed).
- Finding all indexes of occurrences of characters/words (with loops and conditional logic).
- Using ternary operators versus if-elif-else chains, clarifying their situational use (ternary supports only if-else, not elif).
- Reversing the order of words or characters in a string using iteration from the end and managing edge/spaces.
- Transforming/capitalizing string input based on case, using methods like `capitalize`, `title`, and manual checks.
- Approaches for finding the nth occurrence of a character using built-in methods and loop logic.
- Basic calculator using input-driven conditions.
- Matrix (2D, 3D) generation with nested loops and list comprehensions for assignments, with rationale behind the logic.
- Clarified evaluation of string content (e.g., printable/non-printable, decimal/non-decimal) and the implications of certain methods.

Debugging and Tools

- Addressed a frequent issue with Jupyter notebooks: cells stuck in executing state (indicated by star), how to stop execution and avoid restarting the kernel unless necessary.
- Recommended using Google Colab temporarily if Jupyter malfunctions.

Assignment Strategies

- Reiterated the importance of understanding loop vs. method restrictions for assignments.
- Encouraged students to reason out logic before coding, particularly when methods are restricted.
- Explained effective use of flags, counters, and list management when processing input data.
- Clarified cases for using exception handling when methods could raise errors.
- Provided nuanced guidance on expected output formatting for complex assignment tasks (e.g., word separation, preserving separator in output).

Important Deadlines

- Reminder that all assignments (loop, string, list, etc.) are due by the 22nd (extension from the 20th).
- Urged students to submit topics that are complete and avoid waiting until the final deadline.
- Announced that file uploads and class video recordings are maintained and accessible.

Closing Remarks

- Next class will focus on sets and frozen sets, with further assignments on functions, sets, and dictionaries scheduled for distribution by the 22nd/23rd.
- Class ended with general thanks and reminders to seek clarification for outstanding doubts.